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Farmers' struggles with cold storage systems

Farmers' Struggles and Solutions in Cold Storage Systems: Technical, Economic, and Logistical Perspectives

Quick Reference

Key Findings Table

Challenge Area	Key Struggles	Emerging Solutions	Citations
Technical	Non-uniform cooling, poor design, high energy use, unreliable power, knowledge gaps	Curved corners, PCMs, IoT/AI monitoring, solar/off-grid	1 2 3 4 5 6
Economic	High upfront costs, limited financing, low utilization, high transaction costs	Solar-powered, IoT-enabled storage, cooperatives, subsidies	7 8 9 10 11
Logistical	Poor infrastructure, extended waiting, booking inefficiencies, transport costs	GIS planning, digital booking, shared logistics platforms	12 13 14 15 16
Policy/Training	Fragmented support, limited extension, low digital literacy	Integrated policy, training, blended extension, subsidies	11 17 18 19

Direct Answer

Farmers face multiple struggles with cold storage systems, including technical inefficiencies in traditional designs, high energy consumption, and uneven cooling which lead to produce spoilage. Economically, high upfront costs, limited financing, and high transportation or transaction fees hinder their ability to fully capitalize on the benefits of cold storage. Logistically, poor infrastructure, extended waiting times during peak seasons, and inefficient booking systems limit access, particularly for smallholder farmers in rural areas. However, emerging solutions integrating IoT, AI, renewable energy, and digital platforms are showing promising results in improving storage quality, reducing postharvest losses, increasing income through intertemporal price arbitrage, and enhancing overall farmer resilience. 1 6 7 12

Study Scope

- **Time Period:** 2010–2024
- **Disciplines:** Agricultural engineering, rural economics, logistics, policy studies, information technology
- **Methods:** Meta-analysis, empirical studies, case studies, techno-economic modeling, qualitative interviews

Assumptions & Limitations

- Most findings are based on smallholder and developing country contexts; results may differ in industrialized settings.
- Technical innovations (IoT, AI, PCMs) are often pilot-scale or early-stage; long-term reliability and scalability require further study.
- Economic analyses assume typical market conditions; extreme volatility or policy changes may alter outcomes.
- Logistical solutions depend on local infrastructure and digital literacy, which vary widely.
- Policy effectiveness is contingent on implementation quality and institutional capacity.

Suggested Further Research

- Long-term performance and reliability of AI-driven predictive control in rural cold storage.
- Socio-economic impacts of digital platforms on farmer decision-making and market access.
- Optimization of PCM selection and integration for region-specific cold storage needs.
- Comparative studies of cooperative versus private cold storage models in diverse contexts.
- Impact of blended extension services on technology adoption rates.

1. Introduction

Background and Scope

Cold storage is a cornerstone of modern agricultural supply chains, crucial for reducing postharvest losses, maintaining produce quality, and enabling farmers to access better markets and prices. For smallholder farmers, especially in developing countries, cold storage offers the potential to stabilize income, diversify crops, and mitigate risks associated with perishability and market volatility. However, the adoption and effective use of cold storage systems are fraught with technical, economic, and logistical challenges that disproportionately affect rural and resource-constrained communities. This report synthesizes current research to illuminate these struggles and highlight emerging solutions, bridging theoretical foundations with practical insights [12] [20] [21].

2. Theoretical Frameworks

Technical Challenges in Cold Storage Systems

Design and Operational Inefficiencies

- **Traditional cold storage designs** (sharp corners, poor airflow) cause thermal stratification and uneven cooling, leading to spoilage and energy waste [1] [2].

- **Innovative modifications** (curved corners, PCM-buffered backup) improve air circulation, stabilize temperature during compressor downtime, and reduce energy consumption [1].
- **Phase Change Materials (PCMs):** Enhance thermal performance, enable load shifting, and reduce compressor cycling, though require careful selection and integration [22] [23].
- **Advanced control strategies:** Model Predictive Control (MPC) and AI-assisted design optimize PCM-enhanced systems, but real-time optimization and accurate modeling remain challenges [24].

Energy and Environmental Constraints

- **High energy consumption** and unreliable grid electricity limit cold storage adoption, especially for smallholders [4] [25].
- **Environmental impacts:** Refrigerant leakage and high carbon emissions prompt the need for sustainable alternatives (radiative cooling, solar PV) [26].
- **Hybrid solar-battery-PCM systems:** Reduce grid dependence by up to 26%, lower operational costs, and extend temperature retention [27].

Technology Adoption and Knowledge Gaps

- **Limited technical knowledge** among farmers leads to inefficient use and increased losses [5].
- **Extension services:** Farmer Field Schools, Farmer-to-Farmer Extension, and blended learning approaches improve knowledge and adoption, but facility technologies are harder to extend than new varieties [19] [28].
- **On-farm trials and ICT-based extension:** Enhance contextualized learning and technology uptake [29] [30].

Innovations in Cold Storage Design

- **Micro-channel and metal foam structures:** Accelerate heat transfer and reduce freezing time [31] [32].
- **Optimized inlet diffusers:** Improve thermal mixing and energy efficiency [33].
- **IoT/AI-based monitoring:** Real-time environmental sensing, predictive analytics, and automated control improve energy efficiency and reduce spoilage [3] [34] [35].

Synthesis: Technical challenges are multifaceted, spanning design, energy, and knowledge domains. Innovations in materials, control systems, and digital monitoring are promising but require further research and capacity building for widespread adoption [1] [2] [3] [4] [5] [6].

3. Methods & Data Transparency

- **Meta-analysis:** Aggregated findings from empirical studies, case reports, and techno-economic models.

- **Qualitative interviews:** Insights from farmer experiences, extension officers, and cold storage managers.
- **Quantitative modeling:** Energy consumption, cost-benefit analyses, and market impact assessments.
- **Comparative case studies:** Small vs. large cold storage warehouses, solar-powered vs. conventional systems.
- **Data sources:** Peer-reviewed journals, government reports, NGO studies, and pilot project evaluations.

4. Critical Analysis of Findings

Technical Challenges

Comparative Table: Conventional vs. Innovative Cold Storage Designs

Feature	Conventional Design	Innovative Design (Curved/PCM/IoT)	Impact on Farmers
Airflow	Poor, stratified	Uniform, improved mixing	Reduced spoilage
Energy Consumption	High	Lower (up to 28% reduction)	Lower operational costs
Temperature Stability	Fluctuating	Stable (PCM backup, IoT control)	Extended shelf life
Monitoring	Manual, infrequent	Real-time, automated (IoT/AI)	Early spoilage detection
Accessibility	Grid-dependent	Solar/off-grid options	Suitable for rural areas

Analysis: Design innovations directly address technical inefficiencies, but require investment, training, and context-specific adaptation. IoT and AI integration are particularly effective in rural, off-grid settings [2] [3] [27].

Economic Impact

Income, Price Realization, and Market Access

- **Intertemporal price arbitrage:** Cold storage enables farmers to sell produce when prices are higher, increasing net income [7].
- **Crop diversification:** Access to cold storage encourages farmers to grow a wider range of crops, improving resilience [36].
- **Profitability:** Small cold storage warehouses show better utilization and returns than large units, especially for smallholders [11].
- **Solar-powered, IoT-enabled storage:** Reduces losses, increases income by ~22%, and yields ROI of 2–2.1x over three years [6] [8] [37].

Barriers to Adoption and Willingness to Pay

- **High upfront costs, limited credit, and transaction fees** are major barriers [9] [38].
- **Demographic factors:** Gender, education, and training influence willingness to pay; female farmers often show higher WTP [9].
- **Cooperatives:** Collective action enhances economic benefits, facilitates access, and improves income stability [10] [39].

Synthesis: Economic benefits are substantial but unevenly distributed. Targeted subsidies, cooperative models, and training are critical to overcoming adoption barriers [7] [8] [9] [10] [11].

Logistical and Accessibility Issues

Infrastructure and Geographical Barriers

- **Poor rural roads and suboptimal storage locations** increase transport costs and reduce access [13] [14].
- **Geographical disparities:** Rural farmers face longer travel times, higher costs, and greater losses [40].
- **GIS-based planning:** Optimizes storage placement, reduces travel distances, and improves accessibility [16] [41].

Booking Inefficiencies and Peak Season Bottlenecks

- **Extended waiting times and inefficient booking** during peak seasons are caused by limited capacity, poor demand forecasting, and lack of real-time interfaces [12] [15].
- **Digital platforms:** Mobile/web-based booking improves access, reduces waiting, and empowers farmers [12].

Transaction Costs and Market Information Gaps

- **High transaction costs and limited market information** reduce profitability and hinder participation [7] [42].
- **Integrated logistics platforms:** Reduce costs, improve information flow, and enhance cold chain efficiency [43].

Synthesis: Logistical barriers are significant but increasingly addressable through digital platforms, GIS planning, and shared logistics models. Infrastructure investment remains a priority [12] [13] [14] [15] [16].

5. Real-world Implications

Impact on Smallholder Farmers

- **Postharvest loss reduction:** Cold storage reduces losses by up to 40%, directly improving food security and income [8] [37].
- **Income stability:** Enables price arbitrage, crop diversification, and off-farm employment [7] [36].
- **Accessibility:** Solar-powered, IoT-enabled storage is viable for rural, off-grid communities, with payback periods as short as 2.1 years [6] [37].

- **Cooperatives:** Enhance adoption, income, and resilience, especially in contexts of climate shocks and market instability [10] [39].
- **Digital platforms:** Improve booking efficiency, reduce waiting times, and empower farmers with real-time information [12].

Policy and Training

- **Integrated policy frameworks:** Subsidies, credit schemes, cooperative models, and digital platforms are most effective when combined [11] [17].
- **Training and extension:** Farmer Field Schools, blended learning, and ICT-based extension increase adoption and effective use [18] [19].
- **Financial instruments:** Credit and insurance improve adoption rates, especially when paired with training and market access [44].

Synthesis: Real-world impacts are maximized when technical, economic, and logistical solutions are integrated with policy and training interventions. Context-specific strategies are essential for sustainable adoption [8] [11] [17] [18] [19] [37].

6. Future Research Directions

Best Practices

- **Shared storage models:** Enable cost-effective access and reduce seasonal idleness [37] [45].
- **Digital booking platforms:** Real-time updates, location-based search, and simplified reservation processes enhance access [12] [46].
- **Integrated cold chain planning:** GIS and optimization algorithms improve logistics and reduce losses [46].

Emerging Technologies

- **AI, advanced PCMs, digital twins:** Improve spoilage prediction, shelf life, and operational efficiency [24] [47] [48].
- **IoT and renewable energy integration:** Reduce energy costs, improve storage quality, and enable scalable solutions [35] [49].
- **Mobile/web platforms:** Enhance farmer access, information flow, and decision-making [12] [50].

Policy and Institutional Recommendations

- **Integrated interventions:** Combine subsidies, training, cooperative models, and digital platforms for maximum impact [17] [51].

- **Hybrid financing models:** Government support plus community-driven enterprise fosters sustainability [52].

- **Capacity building:** Targeted training, extension, and digital literacy programs are essential [18] [53].

Synthesis: Future research should focus on optimizing advanced control systems, region-specific PCM integration, and socio-economic adoption hurdles. Policy and institutional frameworks must be adaptive, inclusive, and context-sensitive to support sustainable cold storage adoption [17] [24] [47] [48] [51] [52].

Conclusion

Cold storage systems are pivotal for reducing postharvest losses and improving farmer livelihoods, but technical, economic, and logistical challenges persist, especially for smallholder farmers in developing regions. Innovations in design, energy, and digital platforms are promising, but require integrated policy, training, and financial support for sustainable adoption. Future research and practice should prioritize context-specific solutions, capacity building, and collaborative models to fully leverage cold storage technology for resilient agricultural systems [1] [6] [7] [12].

References

1. Design optimization of cold storage for agricultural product using corner curvature and thermal backup system: A case study on brinjal and tomato Swami, M.S., Dahake, M., Deshmukh, M., (...), Ladekar, C. Indian Journal of Engineering and Materials Sciences, 2026 https://www.scopus.com/pages/publications/105047570545?origin=scopusAI

2. Design and Research of a New Cold Storage: The Phase-Temperature Storage (PTS) to Reduce Evaporator Frosting Duan, L., Zheng, Y., Jiang, Y., (...), Li, X. Foods, 2025 https://www.scopus.com/pages/publications/105005099547?origin=scopusAI

3. Smart Cold Storage and Inventory Monitoring System Srivatsa, S.G., Bharadwaj, K.R.M., Alamuri, S.L., (...), Shreenidhi, H.S. 2021 6th International Conference on Recent Trends on Electronics, Information, Communication and Technology, RTEICT 2021, 2021 https://www.scopus.com/pages/publications/85118922067?origin=scopusAI

4. Decentralized solar-powered cooling systems for fresh fruit and vegetables to reduce post-harvest losses in developing regions: A review Amjad, W., Munir, A., Akram, F., (...), Mahmood, F. Clean Energy, 2023 https://www.scopus.com/pages/publications/85163737746?origin=scopusAI

5. Unpacking the harvest losses challenges to enhancing food security among farming communities in Southern Highland, Tanzania Gwambene, B. International Journal of Postharvest Technology and Innovation, 2026 https://www.scopus.com/pages/publications/105041370169?origin=scopusAI

6. An AI-Integrated Renewable-Powered Cold Storage System with Advanced Environmental Sensing for Smallholder Farmers Gowda, S.D., Raj, K.T. Engineering, Technology and Applied Science Research, 2025 https://www.scopus.com/pages/publications/105027906187?origin=scopusAI

7. COST BENEFIT ANALYSIS OF USING COLD STORAGE FACILITIES: EVIDENCE FROM ONION FARMERS IN BONGABON AND SAN JOSE CITY, NUEVA ECIJA, PHILIPPINES Buenaseda, R.D.Q., Daloonpate, A. Journal of the International Society for Southeast Asian Agricultural Sciences, 2024 https://www.scopus.com/pages/publications/85211597868?origin=scopusAI

8. A Techno-Economic Framework for Energy-Efficient and Intelligent Cold Storage Management in Agricultural Supply Chains Gowda, S.D., Raj, K.T. Engineering, Technology and Applied Science Research, 2025 https://www.scopus.com/pages/publications/105026167083?origin=scopusAI

9. To sell or store? Assessing smallholder farmers' willingness to pay for potato cold storage in Kenya Ateka, J.M., Mbeche, R. Food and Energy Security, 2023 https://www.scopus.com/pages/publications/85144079472?origin=scopusAI

10. Sustainability of Philippine carabao center and primary cooperative partnership in carabao-based enterprise Del Rosario, W.T., Vargas, D.S. Buffalo Bulletin, 2013 https://www.scopus.com/pages/publications/84897869006?origin=scopusAI

11. Cold storage warehousing as a risk mitigation strategy in the chilli value chain of Guntur, India Andukuri, R.S., Mahendran, K., Choyal, B. Discover Applied Sciences, 2025 https://www.scopus.com/pages/publications/105024071856?origin=scopusAI

12. Cold Storage Management System Gayathri Devi, Y., Babu, S., Rajyalakshmi, M. Proceedings - IEEE 2024 1st International Conference on Advances in Computing, Communication and Networking, ICAC2N 2024, 2024 https://www.scopus.com/pages/publications/105000635807?origin=scopusAI

13. Advancing Inclusive, Multimodal, Climate-Resilient Planning for Rural Networked Transport Infrastructure Segerberg, B., Noriega, A. Sustainability (Switzerland), 2026 https://www.scopus.com/pages/publications/105034329934?origin=scopusAI

14. Geospatial perspective for post-harvest infrastructure management: Positioning of new cold storage Vijjapu, P., Kimothi, M.M., Roy, S., (...), Ray, S.S. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives, 2019 https://www.scopus.com/pages/publications/85071079696?origin=scopusAI

15. GIS based infrastructure planning of Cold Storages for Crop Preservation: a case study of Kolar District, Karnataka Abhinav, A., Chethan, V., Pancholi, S., (...), Pandey, K. Spatial Information Research, 2022 https://www.scopus.com/pages/publications/85128425309?origin=scopusAI

16. A 3D multiobjective multi-item eco-routing problem for refrigerated fresh products delivery using NSGA-II with hybrid chromosome Thakur, K., Maity, S., Nielsen, P., (...), Maiti, M. Computers and Industrial Engineering, 2024 https://www.scopus.com/pages/publications/85207084443?origin=scopusAI

17. Policy incentives for smallholder adoption of climate-smart agricultural practices van Asseldonk, M., Girvetz, E., Pamuk, H., (...), Ruben, R. Frontiers in Political Science, 2023 https://www.scopus.com/pages/publications/85161016926?origin=scopusAI

18. Does agricultural training and demonstration matter in technology adoption? The empirical evidence from small rice farmers in Tanzania Mgendi, B.G., Mao, S., Qiao, F. Technology in Society, 2022 https://www.scopus.com/pages/publications/85134303571?origin=scopusAI

19. Impact of farmer field schools on agricultural technology extension—evidence from greenhouse vegetable farms in China Cai, J., Hu, R., Hong, Y. Applied Economics, 2022 https://www.scopus.com/pages/publications/85119885677?origin=scopusAI

20. The role of cold chain logistics in reducing postharvest losses Edo, G.I., Ali, A.B.M., Owheruo, J.O., (...), Fuwape, I.A. Next Energy, 2026 https://www.scopus.com/pages/publications/105041194192?origin=scopusAI

21. Cold Chain Infrastructure in India and Its Future Potential Roy, S., Khandelwal, C., Rajesh Sharma, R. 2025 2nd International Conference on New Frontiers in Communication, Automation, Management and Security, ICCAMS 2025, 2025 https://www.scopus.com/pages/publications/105030037061?origin=scopusAI

22. A Review of Applications and Implementation of Phase Change Materials in Cold Storage Facilities Xing, Z., Pan, Y., Yin, R., Zhou, N. Lecture Notes in Civil Engineering, 2026 https://www.scopus.com/pages/publications/105036745235?origin=scopusAI

23. Phase Change Materials for Cold Thermal Energy Storage applications: A critical review of conventional materials and the potential of bio-based alternatives Ouaouja, Z., Ousegui, A., Toublanc, C., (...), Havet, M. Journal of Energy Storage, 2025 https://www.scopus.com/pages/publications/85214585727?origin=scopusAI

24. Towards high-performance cold thermal energy storage with phase change materials: From material modification to structural topology optimization and system application Zhang, D., Wang, Y., Zhang, C., (...), Qiu, L. Journal of Energy Storage, 2026 https://www.scopus.com/pages/publications/105043854276?origin=scopusAI

25. Parametric Based Techno-Economic Evaluation for a Solar Thermal-PV Integrated Multi-Commodity Storage Facility Shahzaib, M., Moeez, A., Memon, A., Kumar, L. Energy Storage, 2024 https://www.scopus.com/pages/publications/85203000658?origin=scopusAI

26. Radiative Cooling Technologies in Agri-food Systems: From Material Innovation to Field Deployment—Review Abdullahi, A., Korumilli, T., Rao, K.J. Food and Bioprocess Technology, 2026 https://www.scopus.com/pages/publications/105042635132?origin=scopusAI

27. Development and evaluation of a hybrid solar-battery powered PCM-integrated portable cold storage system for fruits and vegetables Jain, N.K., Paliwal, D., Jain, P. Physica Scripta, 2025 https://www.scopus.com/pages/publications/105033803279?origin=scopusAI

28. Effectiveness of training offered by Ethiopian institute of agricultural research to farmers: The case of Holetta, Melkassa and Debre Zeit agricultural research centres Tesfaye, T., Karippai, R.S., Tesfaye, T. African Journal of Agricultural Research, 2010 https://www.scopus.com/pages/publications/77951680816?origin=scopusAI

29. Farm trials as an effective extension management method for faster varietal knowledge and diffusion Nayak, S., Dwivedi, R. Constructive Discontent in Execution: Creative Approaches to Technology and Management, 2023 https://www.scopus.com/pages/publications/85162210401?origin=scopusAI

30. Usability of E-Extension Technology and Growth of Agricultural Productivity in Southeast Nigeria: ADP Extension Worker's Survey Obianefo, C.A., Obiekwe, N.J., Okoroji, N.O., (...), Osegbue, E.G. Sarhad Journal of Agriculture, 2024 https://www.scopus.com/pages/publications/85200200370?origin=scopusAI

31. Thermophysical Enhancement of Cold Energy Storage Systems Using Metal Foam and Ternary Nanoparticles: A Numerical Investigation Mirparizi, M. Power Engineering and Engineering Thermophysics, 2026 https://www.scopus.com/pages/publications/105048180815?origin=scopusAI

32. Synergistic integration of micro-channels and thermal storage systems for energy efficiency enhancement in heat transfer applications-a critical review Barmavatu, P., Heeraman, J., Das, M.K., (...), Srinivasnaik, M. Journal of Thermal Analysis and Calorimetry, 2026 https://www.scopus.com/pages/publications/105026406117?origin=scopusAI

33. Numerical study of stratification characteristics in horizontal chilled water storage tanks under high-flow rate and low-temperature difference conditions Lyu, S., Zhou, X., Lyu, G., (...), Li, Z. Applied Thermal Engineering, 2025 https://www.scopus.com/pages/publications/105022205019?origin=scopusAI

34. IoT-based Smart Energy Monitoring and Forecasting using Long Short-Term Memory and Cloud Integration Amune, A., Rajurkar, S., Pawar, A., (...), Musale, V. Proceedings of 3rd International Conference on Sustainable Computing and Data Communication Systems, ICSCDS 2025, 2025 https://www.scopus.com/pages/publications/105018472482?origin=scopusAI

35. Development and performance evaluation of solar-powered cold storage system for perishable agricultural products Hussain, G., Sultan, M., Aleem, M., (...), Ahmad, S. International Communications in Heat and Mass Transfer, 2025 https://www.scopus.com/pages/publications/105011735887?origin=scopusAI

36. A Comparative Study of the Extent of Crop Diversification and Price Realization among Users and Non-Users of Cold Storage Farm Households in Assam Patowary, S. Indian Journal of Economics and Development, 2023 https://www.scopus.com/pages/publications/85164905572?origin=scopusAI

37. Design of a low cost, smart and stand-alone PV cold storage system using a domestic split air conditioner Mishra, R., Chaulya, S.K., Prasad, G.M., (...), Banerjee, G. Journal of Stored Products Research, 2020 https://www.scopus.com/pages/publications/85091582519?origin=scopusAI

38. Factors determining smallholder farmers' willingness to pay for a metal silo in Zimbabwe Chuma, T., Mudhara, M., Govereh, J. Agrekon, 2020 https://www.scopus.com/pages/publications/85078466821?origin=scopusAI

39. Collective action under risk and uncertainty: assessing impacts on smallholder farmers' income and food security in Malawi Suh, J., Ji, S., Nindi-Chigwe, T.C. Frontiers in Sustainable Food Systems, 2025 https://www.scopus.com/pages/publications/105011939545?origin=scopusAI

40. Effects of Postharvest Transportation and Storage on Fruits and Vegetables in Bong County, Liberia Tokpah, D.P., Isbilen, O., Gbelee, F., Page, Z. Research on World Agricultural Economy, 2026 https://www.scopus.com/pages/publications/105048644696?origin=scopusAI

41. Optimizing vehicle utilization in the cold chain: Litereture review Pereglin, T., Stankovic, R. Transportation Research Procedia, 2023 https://www.scopus.com/pages/publications/85184961739?origin=scopusAI

42. Transaction costs magnitudes, market participation, and smallholder profitability in rural-urban vegetable supply chain Baraka, B., Mburu, J., Muriithi, B. International Journal of Vegetable Science, 2021 https://www.scopus.com/pages/publications/85076425634?origin=scopusAI

43. Postharvest handling and storage technologies for fresh horticultural produce Acedo, A.L. Acta Horticulturae, 2019 https://www.scopus.com/pages/publications/85112720192?origin=scopusAI

44. The Impact of Credit and Insurance on Farmers' Climate-Smart Agricultural Technologies Adoption: Evidence from Climatic Transition Zone in China Zhang, B., Xu, W., Yang, P., Ding, K. Sustainability (Switzerland), 2026 https://www.scopus.com/pages/publications/105048784828?origin=scopusAI

45. Research on the construction mode of storage and preservation of cold-chain facilities for agricultural products (fruits and vegetables) Zhang, R., Liu, Q., Liu, B., (...), Chang, Z. Journal of Chinese Agricultural Mechanization, 2024 https://www.scopus.com/pages/publications/85211487324?origin=scopusAI

46. Integrated site selection framework for origin-based cold storage using GIS-MCDM and improved Harris Hawks optimization Li, Y., Li, F., Yang, X., (...), Han, J. Scientific Reports, 2026 https://www.scopus.com/pages/publications/105035347879?origin=scopusAI

47. Digital twin integration for dynamic quality loss control in fruit supply chains Zou, Y., Wu, J., Meng, X., (...), Manzardo, A. Journal of Food Engineering, 2025 https://www.scopus.com/pages/publications/105000565294?origin=scopusAI

48. Bi-objective optimization of hybrid renewable energy systems with ice storage vs. battery storage for sustainable dairy farming Briceño-León, M., Escrivá-Escrivá, G., Clairand, J.-M., Serrano-Guerrero, X. Journal of Energy Storage, 2026 https://www.scopus.com/pages/publications/105033905655?origin=scopusAI

49. IoT-Based Efficient Storage System for Sustainable Agriculture Gajarushi, A., Acharya, A., Banerjee, P., (...), Baghini, M.S. APSCON 2024 - 2024 IEEE Applied Sensing Conference, Proceedings, 2024 https://www.scopus.com/pages/publications/85189938436?origin=scopusAI

50. Farmer Connect Mobile App and Agricultural Equipment E--Business Deore, A.S., Tilekar, V.S., Musale, J., (...), Musale, A.S. Industry 5.0: Innovation and Adoption for Sustainability, 2026 https://www.scopus.com/pages/publications/105049188730?origin=scopusAI

51. Increasing adoption of grain postharvest technology by smallholder farmers: a five-pronged strategy Nakoma-Ngoma, T., Leslie, J.F., Monjerezi, M., (...), Matumba, L. Frontiers in Sustainable Food Systems, 2025 https://www.scopus.com/pages/publications/105015206976?origin=scopusAI

52. Techno-Economic Feasibility of Solar-Powered Cold Storage for Coastal Fisheries Aryanto, N., Giriantari, I.A.D., Satya Kumara, I.N., Setiawan, I.N. Proceedings - International Conference on Smart-Green Technology in Electrical and Information Systems, ICSGTEIS, 2025 https://www.scopus.com/pages/publications/105031658529?origin=scopusAI

53. Cocoa Farmers' Preference and Willingness to Pay for Climate-Smart Extension Services (CES): A Choice Experiment Approach Inkoom, E.W., Dadzie, S.K.N. SAGE Open, 2025 https://www.scopus.com/pages/publications/105004441068?origin=scopusAI